



Green synthesis of titanium dioxide nanoparticles using *Laurus nobilis* (bay leaf): antioxidant and antimicrobial activities

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Abstract

Amalgamation of nanoparticles using “Green Technology” is an intensive field of research due to the increase in the need for the development of new antimicrobial materials. This method is meritorious both in environmental and economic aspects. In the present study, synthesis of titanium dioxide nanoparticles (TiO₂ NPs) was done using the aqueous extract of *Laurus nobilis* (bay leaf). The characterization was done using UV–visible absorption spectroscopy, X-ray diffraction (XRD), Fourier transform infrared spectroscopy (FTIR), scanning electron microscopy (SEM), particle size, and zeta potential analyses. Successful production of nanoparticles was confirmed by the presence of spherical-shaped TiO₂ NPs with the size of ~ 100 nm. The nanoparticles were tested against pathogens to determine their antibacterial and antifungal activities. The antimicrobial and antifungal activities with various concentrations for the generated TiO₂ NPs were investigated and the results showed that the synthesised nanoparticles manifested a good inhibitory activity against both the tested bacteria and the fungal strain. *Bacillus subtilis* and *Aspergillus niger* showed better zone of inhibition against nanoparticle. The observation is that the inhibition zone increases proportionately with the increased nanoparticles concentration. The as-prepared nanoparticles were also subjected to free radical scavenging ability using DPPH and hydrogen peroxide assays. TiO₂ NPs revealed a strong antioxidant activity when compared with ascorbic acid standard. Current study concludes that the concocted TiO₂ NPs, using *L. nobilis* bay leaf, can be scaled up for large-scale industrial production, which would be eco-friendly, economic with a high productivity.

Keywords Bay leaves · *Laurus nobilis* · Antioxidant · Antibacterial agent · Antifungal activity

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Introduction

Nanotechnology (Feynman 1960) is one of the fastest emerging scientific fields, which is associated with nano-meter sized objects. Nanoparticle (NP) is defined as a particle that has all its magnitude less than or equal to 100 nm and may be referred as a nano-object. Properties of nanoparticles are depends on their morphological characteristics (size, shape, and distribution), and these characteristics make them easy and effective interaction with microbes, plants and animals (Siddiqi et al. 2018). They exhibit unique properties such as huge surface area to volume ratio (Nath and Barbhuiya 2014), surface plasmon resonance (SPR), and surface enhance Raman scattering (SERS), which enabled the production of nanoparticles using metal in the recent years. Metal NPs are made of metal precursors (Khan et al. 2019). The unique optical and physical properties of metal nanoparticles led to broad range of applications in various fields

like biomedicine, biophysics, agriculture, cosmetics, energy, environment, food and textile (Salata 2004; Kaliprasad and Narayana 2018; Pestovsky and Martínez-Antonio 2017; Badnore et al. 2019; Wan et al. 2015; Imran et al. 2014; Singh et al. 2017; Sawhney et al. 2008).

There are different approaches by which nanoparticles can be synthesized viz., chemical, physical, and biological. The predominantly using physical method was evaporation–condensation and laser ablation. Physical method in nanoparticle synthesis has several disadvantages, like using tube furnace at atmospheric pressure, large space for tube furnace, consumes higher amount of energy, and time consuming method (Kruis et al. 2000; Magnusson et al. 1999). Although large quantities of nanoparticles can be produced in a shorter time using the chemical method, the capping agents used for size stabilization of the nanoparticles are toxic and non-eco-friendly by-products are generated. Biological approaches mitigate this environmental issue and produce nanoparticles as by-products. Thus, there is an increasing demand for “green nanotechnology” (Krithiga et al. 2015). Biological approaches include the production of nanoparticles using yeast, bacteria, fungi, algae, plants, viruses, and actinomycetes (Mukherjee et al. 2001; Nadaroglu et al. 2017) evolving recently at both the intracellular and extracellular levels (Sharma et al. 2019).

Nanoparticles synthesised using plant are more advantageous over microbial mediated as they are free from toxic chemicals, provide natural capping agents and can reduce the price for culture media preparation and microbial isolation (Singhal et al. 2011). Among the metallic NPs, TiO₂-oriented nanomaterials are of keen area of researches due to their specific and unique properties such as solidity, chemical potent, perpetual electronic band structure, huge surface area, and high quantum efficiency (Lai et al. 2015). Titanium dioxide nanoparticles (TiO₂ NPs) are inorganic non-metallic nanomaterials widely used in cosmetics as sun screens, solar panels, plastic and rubber industries, waste water treatment, construction, paper industry, medical devices, gas sensing, sericulture, agriculture and remediation of dyes, pesticides and antibiotics (Dréno et al. 2019; Sofyan et al. 2017; Ijadpanah-Saravy et al. 2014; Tian et al. 2016; Mattiello and Marchiol 2017; Rodríguez-González et al. 2019; Jafari et al. 2020; Mohajerani et al. 2019; Zand et al. 2020; Yadav et al. 2017). Metal oxide nanoparticles such as TiO₂ possesses a strong antimicrobial activity (Gong et al. 2007; Allahverdiyev et al. 2011). This property increases the demand for the synthesis of TiO₂ NPs employing the most reliable methods. Bacterial strains acquire resistance to the antibiotics and this has raised a serious public health issue for the need to develop new bactericidal materials.

Laurus nobilis L., frequently referred with its common name bay leaf which is native of Mediterranean south region. It is an enduring perennial shrub which belongs to

the family *Lauraceae*. Ever since thousands of years this leaf has applications as an key ingredient for cooking to enhance the food flavour, conventional medicine and essential oil (Batoool et al. 2020). Bay leaf has a astringent taste. The aroma and fragrance of the essential oils that is generated from leaves is different with that produced from other part of the plant. The chemical composition present in bay leaves was citric acid, carbohydrate, flavonoids, eugenol, tannins, steroids, alkaloids and triterpenoids. Bay leaf displays various biologic functions such as wound healing (Nayak et al. 2006), antioxidant (Elmastaş et al. 2006), antimicrobial (Ozcan et al. 2010), antimutagenic (Samejima et al. 1998), and anti-inflammatory activities (Kozan et al. 2006). It was discovered that bay leaf possess antioxidant properties due to the presence of phenolic compounds.

Hence, an attempt has been made in the current investigation to synthesize TiO₂ NPs using aqueous leaf extract of *L. nobilis*, and to investigate the intensified functions of synthesised nanoparticles. The produced TiO₂ NPs were characterized with ultra violet–visible spectroscopy, Fourier transform infrared (FTIR) spectroscopy, X-ray diffraction (XRD), scanning electron microscopy (SEM), particle size, and zeta potential analyses. The antioxidant properties were assayed for their free radical scavenging ability using DPPH and hydrogen peroxide. The antimicrobial potential was characterized for the produced titanium dioxide nanoparticles against bacterial (Gram positive and negative) and fungal strains.

Materials and methods

Chemical and reagents

Titanium tetra isopropoxide (TTIP) was purchased from Sisco Research Laboratories (SRL), India. Ultrapure deionised water was used for the preparation of solutions throughout the study. Additional chemicals used in this work are of pure scientific grade and purchased from Himedia (India). All the chemicals were used as received.

Preparation of bay leaf extract

The healthy and dry leaves of *L. nobilis* were purchased from the local market, Tiruchirappalli, Tamil Nadu, India. To remove impurities, cleansed the leaves with double distilled water and sun dried for a few days. The dried leaves were then powdered and stored in a container for further use. About 20 g of the powdered sample was taken along with 150 mL of deionised water and boiled. Strain the suspension with Whatman no.1 filter paper. For the production of TiO₂ NPs, the end product was collected and stored at 4 °C.

Production of titanium dioxide nanoparticles

The production of TiO₂ NPs was as per the method of Santhoshkumar et al. (2014). In an Erlenmeyer flask containing 80 mL of TTIP (0.1 M), 20 mL of leaf extract was added and let to react under constant stirring for 24 h. After the reaction of *L. nobilis* extract with TTIP, the synthesized nanoparticles turned light green in colour. Finally, the obtained nanoparticles were stored at 4 °C for further characterization.

Physicochemical characterization of TiO₂ NPs

The synthesised nanoparticles were put through various physicochemical representation techniques. The initial confirmation of the successful synthesis of nanoparticles was done by UV–visible absorption spectra obtained using a UV visible spectrophotometer Varian, Carry 5000). FTIR (Thermo Nicolet, Avatar 370) analysis for the TiO₂ NPs was done in a range of 400–4000 cm^{−1} to ascertain the functional groups present. XRD analysis was used to confirm the purity and chemical composition of the as-prepared TiO₂ NPs. XRD analysis was done at an examining rate of 5° min^{−1} with Cu Kα radiation in the range of 2θ = 20°–80° (Bruker AXS Advance powder X-ray diffractometer). SEM (JEOL Model JSM–6390LV instrument) was used to study the structural morphology of the produced titanium dioxide nanoparticles. The synthesised titanium dioxide nanoparticles were subjected to particle size and Zeta potential measurements with Zetasizer Nano ZS, Malvern Instruments Ltd., UK.

Antioxidant assays

DPPH free radical scavenging assay

DPPH (1,1-diphenyl 2-picryl hydrazine) radical scavenging capacity of the titanium dioxide nanoparticles was assessed using the procedure of Singh et al. (2002). 200 μmol of DPPH was prepared in methanol. Nanoparticles were prepared with five concentrations, i.e., 25, 50, 75, 100 and 200 μg mL^{−1}. 2 mL of each DPPH and sample solution were mixed well followed by incubation at ambient temperature for about 30 min in a dark room. The absorbance of the reaction mixture was measured spectrophotometrically at 517 nm. Using methanol as a blank solution absorbance was read at 517 nm. The percentage of scavenging activity was found as follows. The percentage of DPPH radical was determined as follows:

Percentage (%) inhibition of DPPH radical scavenging assay

$$= [(A_0 - A_1)/A_0] \times 100,$$

where A_0 was the absorbance of control, and A_1 was the absorbance of the treated sample. For the comparative study

for the antioxidant activity, the standard chemical used is ascorbic acid.

Hydrogen peroxide scavenging activity

Hydrogen peroxide scavenging activity of the titanium dioxide nanoparticles was determined by monitoring through reduction of H₂O₂ (Ali et al. 2020). Typically, in this experiment, 0.4 mL of produced titanium dioxide nanoparticle solution was mixed with 0.6 mL of 40 mM H₂O₂ solution and the solution was made up to 2 mL with the help of 50 mM sodium phosphate buffer (pH 7.4) and incubated for 40 min at 30 °C. The absorbance recorded at 230 nm. The inhibition percentage for H₂O₂ was calculated using formula as mentioned in the DPPH radical scavenging assay. Ascorbic acid was used as standard.

Antimicrobial activity

Antimicrobial property for the synthesized titanium dioxide nanoparticles was bio assayed with the well diffusion method. Using Gram-positive bacteria such as *Bacillus subtilis*, *Staphylococcus aureus* and Gram-negative bacteria such as *Escherichia coli* and *Klebsiella pneumoniae*. Antifungal activity was tested against *Aspergillus niger*. The control used for bacteria and fungus were tetracycline and flucanazole, respectively. In this method, 20 mL of Mueller–Hinton Agar (MHA, HiMedia, India) was discharged into petri dishes and allowed to solidify. Later, sterile discs of 6 mm thickness were placed on Petri dishes. For the test different concentrations of titanium dioxide nanoparticles (50, 100 and 200 μg/mL) were impregnated onto each disc. Finally, the plates were incubated at 37 °C overnight for bacteria and 25 °C for 5 days for fungus, respectively. The zone of inhibition was measured. The procedure was repeated thrice for attaining reproducibility.

Results and discussion

UV–visible spectroscopy

The synthesized TiO₂ NPs were observed using UV–vis spectroscopy. The conversion of Titanium tetra isopropoxide into TiO₂ nanoparticles can be identified by the formation of clouded white colloidal solution. The colloidal solution was read at absorption spectrum between 200 and 1100 nm (Fig. 1). The maximum absorption peak was observed at 393.80 nm and this confirmed the formation of nanoparticles. Similar findings have been reported previously. Physical characterization were done using FTIR, XRD and SEM, particle size and zeta potential analyses.

Fig. 1 Absorption spectra of TiO₂ NPs using UV–visible spectra

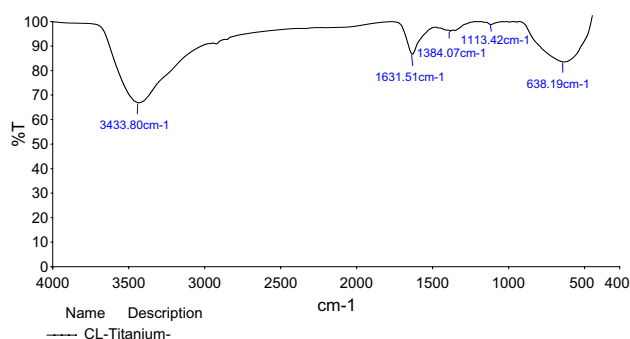
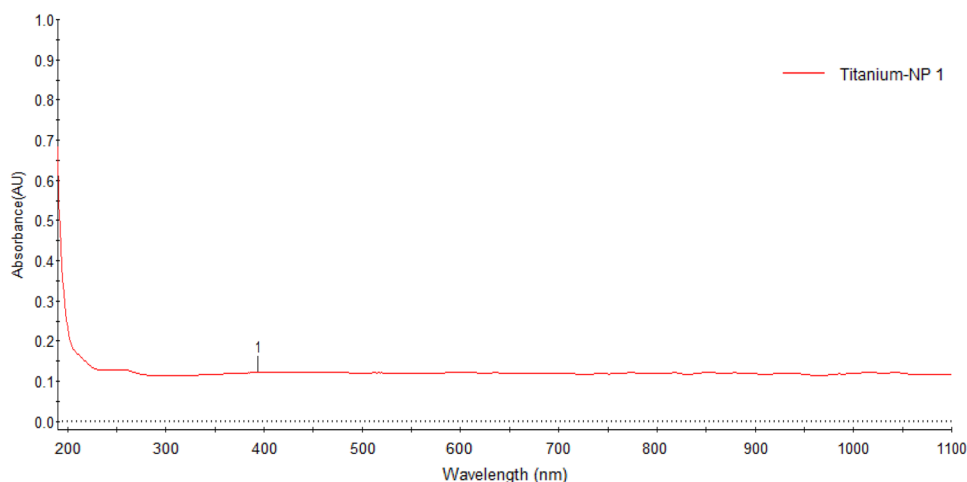


Fig. 2 Fourier transform infrared spectrum of the synthesized TiO₂ NPs

Fourier transform infrared spectroscopy

The chemical composition of the synthesised nanoparticles was analysed using FTIR. At the range 400–4000 cm⁻¹, the spectrum readings were recorded FTIR spectrum showed peaks at 3433.8 cm⁻¹, 1631.5 cm⁻¹, 1384.07 cm⁻¹, 1113.42 cm⁻¹ and 638.19 cm⁻¹ (Fig. 2). The spire at 1384.07 cm⁻¹ indicated C–H rock alkenes. The intense peak centered at 3433.8 cm⁻¹ represented the presence of crystallographic H₂O molecules i.e., O–H stretching bond. The band at around 638.19 cm⁻¹ was corresponding to the O–Ti–O skeletal frequency (Sharfudeen et al. 2017).

Dynamic light scattering (DLS)

The synthesised nanoparticles were subjected to measurements like DLS particle size and zeta potential. For TiO₂ nanoparticles, the primary size found to be approximately 100 nm, the hydrodynamic diameter was found to be 441 nm as measured by DLS (Fig. 3). According to Azouri et al.

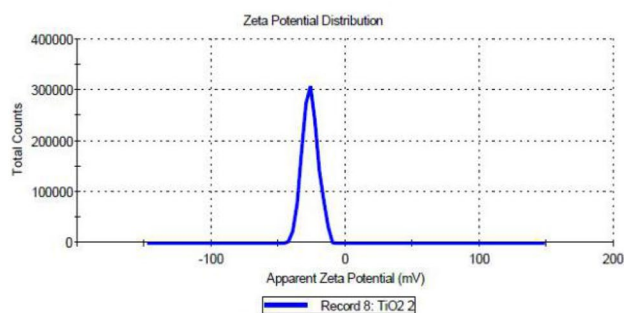
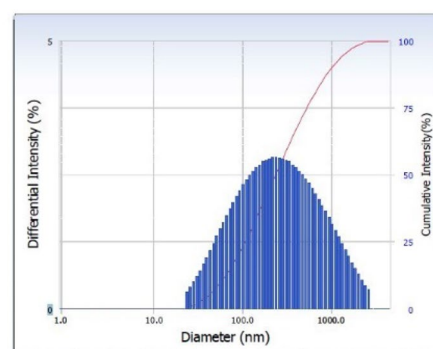


Fig. 3 DLS analysis: **a** particle size, **b** zeta potential of the synthesized TiO₂ NPs

(2006), the suspended colloidal system will be unstable if it has its zeta potential in the range of +25 to –25 mV, and stable if the potential is greater than +25 mV or smaller than –25 mV. It was observed that the zeta potential value for TiO₂ NPs was –26 mV (PDI), which implied a good colloidal nature of the particles that were fairly stable. The results suggested that the synthesised TiO₂ NPs were stable and nano sized, which were consistent with the reported research works (Sethy et al. 2020).

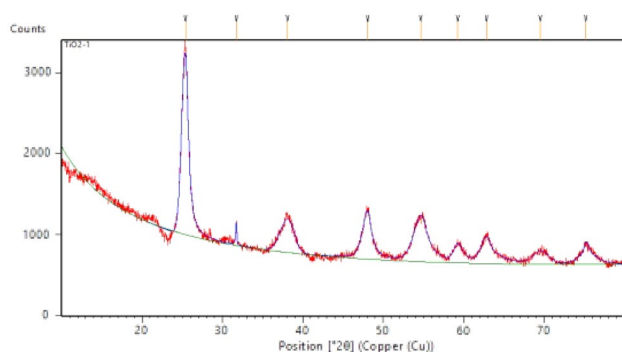


Fig. 4 X-ray diffraction spectrum of synthesized TiO₂ NPs

X-ray diffraction analysis

For the produced nanoparticle, the crystallite size was calculated using XRD. XRD is of great importance in the microstructure characterization of complex, multiphase and single phase materials. The XRD pattern of the TiO₂ sample (Fig. 4) shows the presence of prominent peaks at $2\theta = 25.38, 31.72, 38.10$ and 54.69 corresponding to the planes (110, 101, 200, 211), in accordance with JCPDS data (File No. 99-101-0954). Hence, the synthesis of nanoparticles was confirmed. The crystalline size of the synthesised nanoparticle was found to be in the range of 47–50 nm and the interplanar spacing between atoms was in the range of 0.35–0.15 nm. It has been reported previously that the absence of spurious diffractions would indicate the crystallographic purity (Varshney et al. 2010). There was no spurious diffraction peak found in the sample (Fig. 4).

Scanning electron microscopy (SEM)

Morphological features of the TiO₂ NPs were investigated using SEM. There was a formation of aggregated nanoparticles with the narrow range of dispersion of nanoparticles on the surface. The surface was observed as uneven and the particles were of nano size Fig. 5 shows consistency in the size of nanoparticles at different magnifications. The produced nanoparticles were of size range from 80 to 120 nm with the individual size of 101 nm. The outcome indicated that these surface planes energetically similar to each other though they have different packing structures. The TiO₂ NPs produced using *L. nobilis* leaf extract were globular and rough. The produced nanoparticles have aggregated structures with considerable surface roughness. Similar results were obtained during the study of TiO₂ synthesis mediated by *Echinacea purpurea* (Dobrucka 2017).

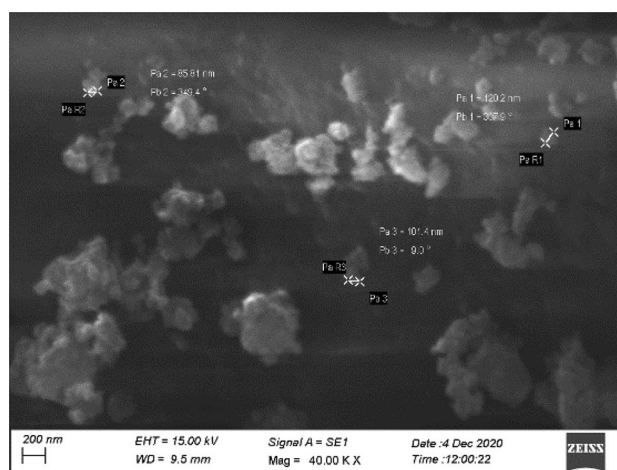


Fig. 5 Scanning electron microscopic image of TiO₂ nanoparticles

Antioxidant activity

Free radicals are produced in human body by numerous endogenous systems, when subjected to various environmental circumstance, physiochemical conditions or pathological states. For normal physiological function, a balance between free radicals and antioxidants is necessary. When the free radicals are left without neutralisation, they are capable of triggering a numeral of human diseases (Lobo et al. 2010). Antioxidants possess strong radical scavenging activities, which is of great importance in the prevention of numerous diseases including cancer, cardiovascular dysfunction, hypertension, cerebrovascular accidents, heart failure, reproductive dysfunction, septic shock, aging, atherosclerosis, diabetes mellitus, rheumatoid arthritis and neurodegenerative disease (Anas et al. 2010; Aviram 2000; Halliwell 2001; Hitchon and El-Gabalawy 2004; Klaunig and Kamendulis 2004; Nunomura et al. 2006; Rattan 2006; Santos et al. 2006; Stocker and Keaney 2004; Szabo 2009; Wood-Kaczmar et al. 2006).

Free radical scavenging ability of the TiO₂ NPs was tested by DPPH radical scavenging assay. The synthesized TiO₂ NPs exhibited an effective inhibitory activity on DPPH radicals when compared with the ascorbic acid standard. The DPPH scavenging activity of the nanoparticles was the maximum of 46.71% at 200 µg/mL and found to be increasing with respect to an increased concentration of TiO₂ NPs (Fig. 6). The results showed that the nanoparticles have DPPH radical scavenging activity depends with respect to the concentration of nanoparticles. There was no significance difference between the ascorbic acid standard and the

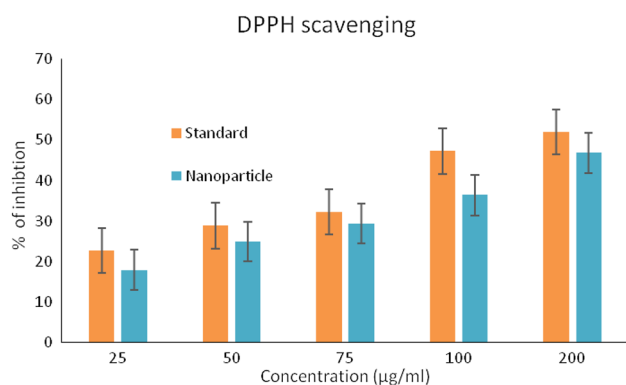


Fig. 6 Antioxidant activity of the synthesised TiO₂ NPs on DPPH radicals

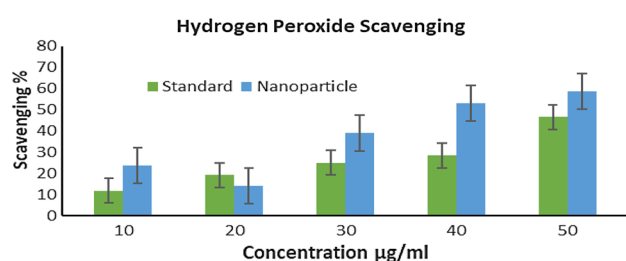


Fig. 7 Antioxidant activity of the synthesised TiO₂ NPs on hydrogen peroxide radicals

nanoparticles and these results were almost similar to those observed earlier (Santhoshkumar et al. 2014).

The ability of the titanium dioxide nanoparticle to scavenge hydrogen peroxide was determined. Alike DPPH assay, in hydrogen peroxide assay, the synthesised TiO₂ NPs exhibited better scavenging ability with the maximum activity of 58.45% at 50 µg/mL and the percentages of inhibition at respective concentrations were similar to the standard antioxidant ascorbic acid, mostly, at the highest concentration as shown in Fig. 7. In an earlier report, TiO₂ NPs revealed 57–72% scavenging activity at 100 µg/mL concentration, however, in 10 µg/mL, scavenging was only 4–9% demonstrating inability to inhibit at lower concentrations (Ajmal et al. 2019). Thus, the results of the present study were in concordance with earlier studies.

Antimicrobial potential

Nanoparticles and their ions generate free radicals and this lead to the development of induced oxidative stress (i.e., ROS, reactive oxygen species). The generated ROS are capable of damaging and destroying the cellular components of the microorganisms in an irreversibly way, resulting in cell death as reported by Khezerlou et al. (2018). The antimicrobial activity of the fabricated TiO₂ NPs were investigated

Table 1 Antimicrobial activity of titanium dioxide nanoparticles against selected strains

Organism	PC	NC	50 (µg/ml)	100 (µg/ml)	200 (µg/ml)
<i>Aspergillus niger</i>	14	–	13	14	14
<i>Klebsiella pneumoniae</i>	14	–	12	13	13
<i>Escherichia coli</i>	15	–	14	15	15
<i>Bacillus subtilis</i>	16	–	15	16	16
<i>Staphylococcus aureus</i>	14	–	12	13	13

PC positive control (standard antibiotic), NC negative control (water)

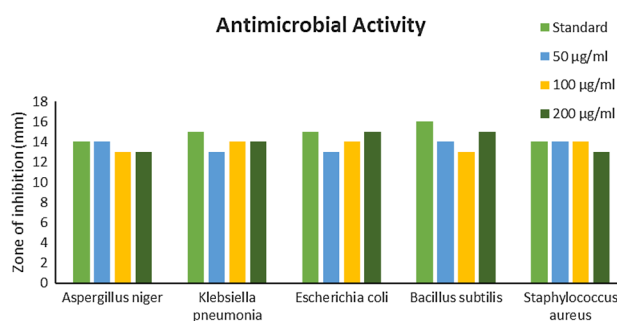


Fig. 8 Antimicrobial activity at various concentrations of titanium dioxide nanoparticles

by the well diffusion method and depicted in Table 1 and in Fig. 8. It was found that TiO₂ NPs shows superior antibacterial activity against all Gram-positive and -negative bacteria. Better activities were observed specifically with *Bacillus subtilis* (Fig. 8). The inhibition zone against *Bacillus subtilis* was in the range of 10–16 mm with TiO₂ nanoparticles synthesised using *L. nobilis* leaf extract when tested at a concentration of 50–100 µg/mL. The zone of inhibition against the fungus, *Aspergillus niger*, was in the range of 10–14 mm with TiO₂ NPs when tested at 50–100 µg/mL concentration. The results showed the TiO₂ NPs to be potent against both Gram-positive and Gram-negative bacteria. The synthesized TiO₂ NPs also displayed potential antifungal activity. These findings corroborated well with the earlier reports (Ajmal et al. 2019; Madasamy et al. 2020).

Conclusion

The present study has suggested a novel method (mentioned) that has been capable of reducing TIIP to TiO₂ NPs using *Laurus nobilis* leaf extract. The results also suggested that the synthesised TiO₂ NPs were spherical, crystalline, 80–120 nm in size and highly pure. The synthesised nanoparticles exhibited significant antioxidant potential and antimicrobial activity. TiO₂ NPs have a broad range of

applications in the field of medicine due to their antibacterial and antifungal properties. Thus, the reported nanoparticle synthesis can be scaled up for large-scale industrial production, which would be of low cost and eco-friendly as they have a wide range of medical applications beneficial to human beings.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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